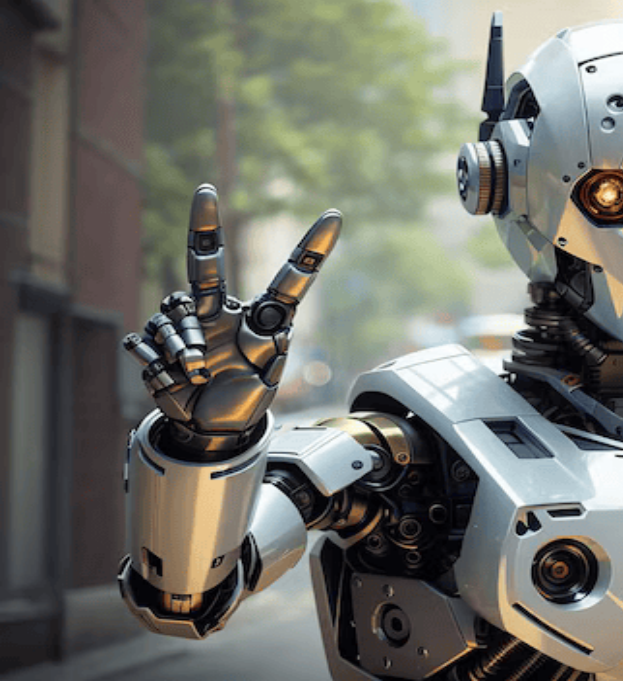




AI for SARS

AI Co-Creation, Prototyping, and Pitching

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GRIT Lab Africa

DAY II



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Pitch Preparation & Solution Pitching

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Day 1 Recap

Day 1 Summary — Key Takeaways



- **AI Fundamentals:** What is AI and what is not, 10 levels of 4IR innovations, machine learning, AI techniques, and applied AI technologies (deep fakes, apps from GRIT Lab Africa).

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- **SARS Landscape:** Technology systems, AI initiatives, and institutional context.

Day 1 Summary — Key Takeaways



- **AI Fundamentals:** What is AI and what is not, 10 levels of 4IR innovations, machine learning, AI techniques, and applied AI technologies (deep fakes, apps from GRIT Lab Africa).
- **SARS Landscape:** Technology systems, AI initiatives, and institutional context.
- **Operational Challenges:** Four critical pain points to address through co-creation.

Day 1: SARS Pain Points



- **ML System Accuracy Issues and Model Validation**

Day 1: SARS Pain Points



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- **Infrastructure & Public Perception**
Taxpayer complaints about digital systems and societal impact.

Day 1: SARS Pain Points



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System slowdowns — *not major*.

Day 1: SARS Pain Points



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Digital audit trails are bypassed.

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AI Toolkit Overview

SARS AI Toolkit (1–2): Education & Digital Assistance



- **1. AI Serious Game for Tax Literacy**
 - Gamified AI scenarios by taxpayer type (individual, SME, corporate)
 - Adaptive difficulty + real SARS policy simulation
 - Behavioural nudging for voluntary compliance
 - Solves: low compliance, digital literacy gaps

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- **2. Digital Human Tax Assistant (Multilingual)**
 - Conversational AI avatars (web, mobile, kiosks)
 - Trained on SARS policies; real-time responses
 - Live translation (isiZulu, isiXhosa, Sesotho, Afrikaans, etc.)
 - Emotion-aware dialogue modelling
 - Solves: accessibility, language barriers, call-centre overload

SARS AI Toolkit (3-4): Segmentation & eFiling Intelligence



- **3. AI Taxpayer Segmentation Engine**
 - Dynamic ML clustering (not static segmentation)
 - Behavioural profiling + compliance risk tiers
 - Predictive intervention triggers
 - Solves: informal economy, underreporting

SARS AI Toolkit (3–4): Segmentation & eFiling Intelligence



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 - Predictive intervention triggers
 - Solves: informal economy, underreporting
- **4. Intelligent eFiling Copilot**
 - Embedded AI assistant within eFiling
 - Plain-language field explanations
 - Real-time error detection + correction suggestions
 - Solves: filing errors, repeated queries, refund delays

SARS AI Toolkit (5–6): Fraud Detection & Explainable Risk



- **5. AI Fraud & Anomaly Detection System**
 - Ensemble ML + Graph AI models
 - Identity theft + refund fraud detection
 - Shell company network discovery
 - Solves: fraud, cyber risk, revenue leakage

SARS AI Toolkit (5–6): Fraud Detection & Explainable Risk



- **5. AI Fraud & Anomaly Detection System**
 - Ensemble ML + Graph AI models
 - Identity theft + refund fraud detection
 - Shell company network discovery
 - Solves: fraud, cyber risk, revenue leakage
- **6. Explainable Risk Scoring Engine (XAI Layer)**
 - Transparent risk models (SHAP/LIME explanations)
 - Full audit trail generator
 - Legal defensibility dashboard
 - Solves: governance, fairness, decision opacity

SARS AI Toolkit (7–8): Process Intelligence & Workflow Automation



- **7. Process Mining & AI Diagnostics Platform**
 - AI-driven process intelligence across SARS workflows
 - Bottleneck detection + throughput / rework insights
 - Decision density measurement + AI opportunity recommendations
 - Solves: legacy inefficiencies, workflow delays

SARS AI Toolkit (7–8): Process Intelligence & Workflow Automation



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 - AI-driven process intelligence across SARS workflows
 - Bottleneck detection + throughput / rework insights
 - Decision density measurement + AI opportunity recommendations
 - Solves: legacy inefficiencies, workflow delays
- **8. Intelligent Case Routing & Workflow Automation**
 - Adaptive case allocation using reinforcement learning
 - Smart queues + exception handling + escalation prediction
 - SLA optimisation via dynamic routing policies
 - Solves: peak overload, uneven resource allocation, backlogs

SARS AI Toolkit (9–10): Predictive Compliance & Policy Simulation



- **9. Predictive Compliance Intervention Engine**
 - Predicts non-compliance risk before it occurs
 - Behavioural AI + targeted nudges (SMS/WhatsApp)
 - Gamified incentives to boost voluntary compliance
 - Solves: low voluntary compliance, late interventions

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 - Predicts non-compliance risk before it occurs
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 - Gamified incentives to boost voluntary compliance
 - Solves: low voluntary compliance, late interventions
- **10. AI Policy Simulation Lab (SARS Process Digital Twin)**
 - Simulates SARS processes to test rules before rollout
 - Scenario modelling + revenue impact forecasting
 - Legal / operational risk modelling for policy changes
 - Solves: frequent legislation changes, rollout risk, rework

SARS AI Toolkit (11-12): Voice AI & Blockchain Verification



- **11. Multilingual Voice AI & IVR Intelligence**
 - Replaces static IVR with natural language voice bot
 - Call sentiment analysis + intent detection
 - Dynamic routing to correct channel / agent
 - Solves: call-centre congestion, low accessibility

SARS AI Toolkit (11-12): Voice AI & Blockchain Verification



- **11. Multilingual Voice AI & IVR Intelligence**
 - Replaces static IVR with natural language voice bot
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 - Solves: call-centre congestion, low accessibility
- **12. Blockchain-Based Refund Verification Layer**
 - Smart-contract validation for high-risk refunds
 - Immutable audit trails + identity integrity checks
 - Cross-agency / cross-system verification workflows
 - Solves: refund fraud, trust deficits, weak traceability

SARS AI Toolkit (13–14): Knowledge Graphs & Dispute Intelligence



- **13. Knowledge Graph of Tax Law & Precedents**
 - Semantic graph linking legislation, rulings, objections
 - Intelligent search + relationship-based retrieval
 - Case similarity matching + precedent referencing
 - Solves: legal complexity, inconsistent interpretation

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 - Semantic graph linking legislation, rulings, objections
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 - Solves: legal complexity, inconsistent interpretation
- **14. AI-Assisted Objections & Dispute Resolution System**
 - NLP case summarisation + document triage
 - Case clustering + probability-of-outcome modelling
 - Settlement suggestion engine to reduce backlogs
 - Solves: dispute backlogs, inconsistent adjudication

SARS AI Toolkit (15–16): Staff Copilots & Revenue Forecasting



- **15. AI Workforce Augmentation Dashboard**
 - Internal AI copilot for PDE and policy specialists
 - SOP drafting assistant + impact analysis generator
 - Decision-support insights for process / policy work
 - Solves: manual analysis burden, slow artefact production

SARS AI Toolkit (15–16): Staff Copilots & Revenue Forecasting



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 - Solves: manual analysis burden, slow artefact production
- **16. Real-Time Revenue Forecasting & Macro AI Model**
 - Time-series forecasting + macroeconomic data fusion
 - Sector volatility modelling + scenario planning
 - Budget planning integration using AI projections
 - Solves: fiscal uncertainty, planning lag, weak foresight

SARS AI Toolkit (17-18): Cyber Defense & VR Tax Education



- **17. Cyber AI Defense Mesh**

- Self-learning cybersecurity using behavioural detection
- Adaptive threat modelling + anomaly-based intrusion detection
- Automated containment and response orchestration
- Solves: cyberattack risk, data breach exposure

SARS AI Toolkit (17–18): Cyber Defense & VR Tax Education



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- **18. AI-Driven Tax Education VR Experience**
 - VR simulation: how tax funds services + infrastructure
 - VR scenarios: consequences of non-compliance
 - Visualises fiscal flows to build understanding and trust
 - Solves: low awareness, weak trust, poor engagement

SARS AI Toolkit (19–20): Trust Analytics & AI Governance



- **19. AI Sentiment & Trust Analytics Engine**
 - NLP across social media, call logs, complaints
 - Public trust index + service quality signals
 - Early warning indicators for reputational risk
 - Solves: reputation risk, service dissatisfaction

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 - NLP across social media, call logs, complaints
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- **20. Enterprise AI Governance & Ethics Platform**
 - Central AI registry + model documentation
 - Bias auditing + fairness scoring + continuous monitoring
 - Compliance mapping + controlled deployment standards
 - Solves: ethical, legal, and reputational AI risk

Co-Creation Studio I: Pain Points v AI-driven Solutions

Problem Mapping — Strategic Pain Point to AI Toolkit



- Pick one or more problems (per group), and map it to a solution from the AI Toolkit.

Solution Architecture — Design Requirements

- **Data Requirements:** Sources, quality, availability, readiness assessment.



Solution Architecture — Design Requirements



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- **Model Selection:** Supervised, unsupervised, hybrid, or rules-based approaches.

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- **System Integration:** Legacy compatibility, APIs, real-time vs. batch processing.

Solution Architecture — Design Requirements



- **Data Requirements:** Sources, quality, availability, readiness assessment.
- **Model Selection:** Supervised, unsupervised, hybrid, or rules-based approaches.
- **System Integration:** Legacy compatibility, APIs, real-time vs. batch processing.
- **Output Layer:** Alerts, risk scores, dashboards, automated recommendations.

Workflow Redesign — Current vs. AI-Enhanced State



- **Current State Mapping:** Manual processes, delays, decision bottlenecks.

Workflow Redesign — Current vs. AI-Enhanced State



- **Current State Mapping:** Manual processes, delays, decision bottlenecks.
- **AI Value Insertion Points:** Where automation or intelligence improves performance.

Workflow Redesign — Current vs. AI-Enhanced State



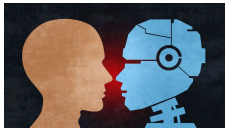
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Workflow Redesign — Current vs. AI-Enhanced State



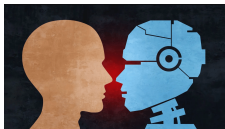
- **Current State Mapping:** Manual processes, delays, decision bottlenecks.
- **AI Value Insertion Points:** Where automation or intelligence improves performance.
- **Future State Design:** Reduced turnaround time and improved compliance outcomes.
- **Estimated Impact:** Efficiency gain, cost reduction, improved accuracy or service delivery.

Human-in-the-Loop — Governance and Oversight



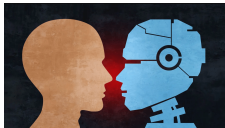
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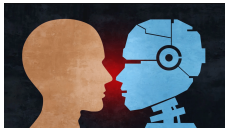
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Human-in-the-Loop — Governance and Oversight



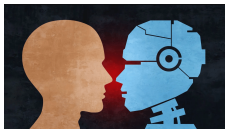
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- **Bias Monitoring:** Mechanisms for detecting unintended discrimination.

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- **Escalation Protocols:** Conditions under which AI decisions are overridden.
- **Bias Monitoring:** Mechanisms for detecting unintended discrimination.
- **Regulatory Compliance:** Audit trails, transparency requirements, POPIA alignment.

Co-Creation Studio II

Next Phase: Evaluation & Strengthening

- **Solution Refinement**

- Tighten design + clarify impact metrics
- Remove non-strategic complexity



Next Phase: Evaluation & Strengthening



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 - Tighten design + clarify impact metrics
 - Remove non-strategic complexity
- **Validation of Assumptions**
 - Test feasibility + readiness
 - Align with SARS operational realities

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 - Tighten design + clarify impact metrics
 - Remove non-strategic complexity
- **Validation of Assumptions**
 - Test feasibility + readiness
 - Align with SARS operational realities
- **Governance & Risk Stress-Testing**
 - Legal, bias, and compliance review
 - Failure-mode + edge-case testing

Pitch Preparation & Solution Pitching

Crafting Your Solution Narrative (1/3)



- **Problem Statement:** Clear articulation of pain point and impact
- **Solution Overview:** AI technology and approach
- **Expected Benefits:** Tangible outcomes and value

Solution Pitching Overview



- **Format:** 5-minute pitch + 2-minute Q&A
- **Evaluation Panel:** SARS leadership and innovation experts
- **Recognition:** Top solutions receive awards and documentation

Reflection & Closing

Your Feedback on Day II — a word or phrase (Closed)

- Use one word, a phrase, or a short sentence to describe Day I.

Thank You — Please Let's Connect

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